

KATHERINE R. GOLD

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University of Arizona, Tucson, AZ

EDUCATION

- Ph.D. Candidate in Chemistry** 2021 – Present
University of Arizona, Tucson, Arizona
Minor: Astrobiology
- A.B. in Mathematics** (magna cum laude) 2017 – 2021
Bryn Mawr College, Bryn Mawr, Pennsylvania
Minors: Physics, Astronomy

RESEARCH APPOINTMENTS

- Graduate Research Assistant** 2021 – Present
Department of Chemistry, University of Arizona
Advisor: Prof. Lucy M. Ziurys
Dissertation (in preparation): “Chemistry in protoplanetary and planetary nebulae: Evidence of nebular shaping and an organic continuum”
- Undergraduate Summer Research Assistant** 2020 – 2021
Department of Physics and Astronomy, Swarthmore College
Research advisor: Prof. Deborah R. Schmidt
Project: “Molecular abundance of HCN and HCO⁺ in planetary nebulae”
- Undergraduate Research Assistant** 2020
Department of Physics and Astronomy, Haverford College
Research advisor: Prof. Karen L. Masters
Project: “Spiral-arm morphology and supermassive black hole mass”

PUBLICATIONS

Statistics: 52 total citations

1. **Gold, K. R.** & Ziurys, L.M. “The Chemistry of Planetary Nebulae: The Ellipticals”, in review at The Astrophysical Journal.
2. **Gold, K. R.** & Ziurys, L.M. “New Detections of the Small Organic Molecules H₂CO and c-C₃H₂ in Planetary Nebulae”, accepted at The Astrophysical Journal on 16 Sep 2026.
3. **Gold, K.R.** & Ziurys, L.M. “Molecular Observations of the Protoplanetary Nebula Frosty Leo: A Unique Window into Superoxygen Chemistry”, 2025, The Astrophysical Journal, 994, 81.
4. Schmidt, D.R., **Gold, K.R.**, Liszt, H., & Ziurys, L.M. “ALMA Observations of CO in Five Planetary Nebulae: Insights into Nebular Shaping”, 2025, The Astrophysical Journal, 992, 98.
5. **Gold, K.R.**, Schmidt, D.R., & Ziurys, L.M. “Expanding the inventory of molecule-rich planetary nebulae: New observations of M4-17, Hu1-1, M1-59, and Na 2”, 2024, The Astrophysical Journal, 976, 196.
6. Koelemay, L.A., **Gold, K.R.**, & Ziurys, L.M. “Phosphorus-bearing molecules PO and PN at the edge of the Galaxy,” 2023, Nature, 623, 292.
7. Schmidt, D.R., **Gold, K.R.**, Sinclair, A., Bergstrom, S., & Ziurys, L.M. “HCN and HCO⁺ in Planetary Nebulae: The Next Level,” 2022, The Astrophysical Journal, 927, 46.

FELLOWSHIPS & COMPETITIVE FUNDING

Kathryn K. and James D. Steele Fellowship, University of Arizona (\$5,000)	2026 – 2027
ARCS Marley Foundation Scholar Award in Memory of Milton "Bud" Webb (\$11,000 + tuition)	2026 – 2027
College of Science Data Science Ambassador Scholarship, University of Arizona (\$1,500)	2025 – 2026
Carl S. Marvel Memorial Fellowship, Dept. of Chemistry & Biochemistry, University of Arizona (\$12,500)	2025
Michael Cusanovich Galileo Circle Scholarship, University of Arizona (\$5,000)	2025
Kathryn K. and James D. Steele Fellowship, University of Arizona (\$10,000)	2024 – 2025
Presidential Merit Scholarship, Bryn Mawr College (\$100,000)	2017 – 2021
Community Service Scholarship, Rotary Club, Chambersburg, PA (\$9,600)	2017 – 2021
Joan Chang Wong 1956 Memorial Internship Fund Award, Bryn Mawr College (\$4,000)	2020

RESEARCH GRANTS

Arizona Astrobiology Center Seed Grant, University of Arizona (\$5,000)	2026
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HONORS & AWARDS

Inaugural Student of the Month, Dept. of Chemistry & Biochemistry, University of Arizona	September 2026
Outstanding Scholarship Award, Dept. of Chemistry & Biochemistry, University of Arizona	2026
Chambliss Astronomy Achievement Student Honorable Mention, American Astronomical Society	2026
Award for Outstanding Excellence in Service, College of Science, University of Arizona	2025
Outstanding Service Award, Dept. of Chemistry & Biochemistry, University of Arizona	2025
Chambliss Astronomy Achievement Student Honorable Mention, American Astronomical Society	2021

TRAVEL GRANTS

Graduate & Professional Student Council Travel Grant, University of Arizona (my half: \$998)	2026
Laboratory Astrophysics Division Travel Grant, American Astronomical Society (\$734)	2025
Graduate & Professional Student Council Travel Grant, University of Arizona (\$827)	2025
Department of Chemistry & Biochemistry Travel Grant, University of Arizona (\$500)	2025
Department of Chemistry & Biochemistry Travel Grant, University of Arizona (\$1,000)	2024
Graduate & Professional Student Council Travel Grant, University of Arizona (\$645)	2024
Astrochemistry Subdivision Symposium Travel Grant, American Chemical Society (\$250)	2023
Department of Chemistry & Biochemistry Travel Grant, University of Arizona (\$1,000)	Fall 2023
Department of Chemistry & Biochemistry Travel Grant, University of Arizona (\$500)	Spring 2023
Department of Chemistry & Biochemistry Travel Grant, University of Arizona (\$500)	2022
Graduate & Professional Student Council Travel Grant, University of Arizona (my half: \$1,242)	2022

SELECT PRESENTATIONS

2026 · Molecular Inventory of the Cotton Candy Protoplanetary Nebula. 79th International Symposium on Molecular Spectroscopy, Urbana-Champaign, IL. **Oral**

2026 · Morphology as a Chemical Predictor: Molecules in Elliptical Planetary Nebulae. 79th International Symposium on Molecular Spectroscopy, Urbana-Champaign, IL. **Oral**

2026 · Millimeter Mapping of CO and HCN in the Dumbbell Nebula: Elucidating Velocity Structure and Chemistry. 248th Meeting of the American Astronomical Society, Pasadena, CA. **Oral**

2026 · Constraining Late-Stage Neutron Capture with H₂S and SiO Isotopologues in O-Rich Protoplanetary Nebulae. AstroNuc 2026 Workshop, Tucson, AZ. **Oral**

2026 · A Survey of H₂S ...: Sulfur Isotopic Anomalies and Neutron Capture. 247th Meeting of the American Astronomical Society, Phoenix, AZ. **Poster — Chambliss Honorable Mention**

2025 · Molecular Complexity in Oxygen-Rich PPNe: New Insights into Sulfur and Silicon Chemistry. Pacificchem 2025, Honolulu, HI. **Oral**

2025 · Formaldehyde and Cyclopropenylidene in Planetary Nebulae: New Observations in M1-7, M1-8, M4-9, and NGC 6445. 78th International Symposium on Molecular Spectroscopy, Urbana-Champaign, IL. **Oral**

2025 · A Plethora of Sulfur-Bearing Molecules in the O-Rich Protoplanetary Nebula Frosty Leo. 246th Meeting of the American Astronomical Society, Anchorage, AK. **Oral**

2025 · Complex Chemistry of Planetary Nebulae: Observations versus Theory. American Chemical Society Spring Meeting, San Diego, CA. **Oral**

2024 · Chemical Complexity in Carbon-Rich Protoplanetary Nebulae: Insights from IRAS Z02229+6208. 244th Meeting of the American Astronomical Society, Madison, WI. **Oral**

2024 · The Organic Molecular Continuum: Bridging the Gap at Planetary Nebulae. American Physical Society March Meeting, Minneapolis, MN. **Oral**

TEACHING

University of Arizona (public R1 land-grant Hispanic-serving institution university)

ASTR/PTYS 206 “Exploring our Solar System” (Lecture), 0.25 FTE TA	Fall 2026
ASTR/PTYS 206 “Exploring our Solar System” (Lecture), 0.50 FTE TA	Spring 2026
ASTR/PTYS/CHEM 488/588 “Astrochemistry”, facilitated Kitt Peak 12m observatory tour	Fall 2025
CHEM 400B “Physical Chemistry Measurements Lab” (Lab & Discussion), 0.50 FTE TA	Spring 2025
CHEM 182 “General Chemistry 2 (Majors) Lab”, 0.25 FTE TA	Spring 2024
CHEM 163 “Honors General Chemical Thinking 1 Lab ”, 0.25 FTE TA	Fall 2023
ASTR/PTYS/CHEM 488/588 “Astrochemistry”, facilitated Kitt Peak 12m observatory tour	Fall 2023
CHEM 144 “Quantitative General Chemistry Lab 2”, 0.25 FTE TA	Spring 2023
CHEM 143 “Quantitative General Chemistry Lab 1”, 0.25 FTE TA	Fall 2022
CHEM 164 “Honors General Chemical Thinking 2 Lab ”, 0.25 FTE TA	Spring 2022
CHEM 162 “Honors General Chemical Thinking 2” (Lecture), 0.25 FTE TA	Spring 2022
CHEM 163 “Honors General Chemical Thinking 1 Lab ”, 0.25 FTE TA	Fall 2021
CHEM 161 “Honors General Chemical Thinking 1” (Lecture), 0.25 FTE TA	Fall 2021

Haverford College (private nonprofit liberal arts college)

ASTR 344 “Multi-wavelength Astronomy & Astrophysics”, TA	Spring 2021
ASTR 101 “Introductory Astronomy”, TA	Spring 2021
ASTR 204 “Introduction to Astrophysics”, TA	Fall 2020
ASTR 101 “Introductory Astronomy”, TA	Spring 2020
PHYS 105 “Electromagnetism”, Grader	Spring 2020
ICPR 246 “Management and Leadership”, Notetaker	Spring 2020

Bryn Mawr College (private nonprofit historically women’s liberal arts college)

MATH 101 “Calculus 1”, Grader	Fall 2020
PHYS 101 (Postbaccalaureate Premedical Program) “Classical Mechanics”, Grader	Fall 2018

Scientific Mentoring:

- Zhiyu Ying ('27) – Undergraduate research (Arizona Astrobiology Center, University of Arizona) 2026
 - Honors thesis on planetary nebulae M4-9 and NGC6445, will prepare 1st author ApJ manuscript
- Matthew Gilles ('25) – Undergraduate research (University of Arizona) 2024 – 2025
 - Supported by \$3,000 from my 2024 Steele Fellowship for summer 2025 research; Honors thesis on N₂H⁺ in 15+ galactic edge molecular clouds, with 1st author paper in review for ApJ
- Vrinda McBride ('27) – Undergraduate research (University of Arizona) 2024 – 2025
 - Comparison of gas and dust in oxygen-rich and carbon-rich planetary nebulae literature review

SERVICE, LEADERSHIP & OUTREACH

AstroCheminar Webinar Series Co-Organizer 2023 – Present

Astrochemistry Subdivision, PHYS Division, American Chemical Society

- Directs monthly online seminars featuring graduate students, post docs, and established researchers in laboratory, experimental, and computational astrochemistry, fostering networking and knowledge exchange

Volunteer 2023 – Present

Program to Advance Women Scientists, University of Arizona

- Demonstration Volunteer at the “Spooktacular Science” Event (2023, 2024, 2025, 2026) and Science City at the Tucson Festival of Books (2024, 2025, 2026)
- Preparatory Volunteer for the ChemCamp for Neurodivergent K–12 Students in collaboration with the Autism Society of Southern Arizona (2024)
- Preparatory Volunteer for Mole Day for National Chemistry Week, American Chemical Society (2025)
- Volunteer for the Chemistry & Biochemistry Graduate Student Recruitment Weekend Social (2024)

Guest Scientist 2024 – Present

Skype a Scientist

- Answered astronomy-related questions for a 6th grade class at Camanche Middle School (Iowa) about stars and space-themed movies (2026)
- Answered astronomy-related questions for a 6th grade class at Barr-Reeve Elementary (Indiana) to supplement their astronomy unit (2026)
- One of three invited speakers of the “Decoding the Universe – Exploring the Unknown with Nature’s Hidden Language” series at Benalla Library (Victoria, Australia) as part of National Science Week (2025)
- Modeled the Skype a Scientist program to high school science teachers at Hillsborough County Public Schools’ (Florida) Professional Development Day to promote classroom adoption (2025)
- Answered astronomy-related questions for 3rd graders at Osseo-Fairchild Elementary School (Wisconsin) to enrich their reading class’s astronomy unit (2025)
- Individually mentored three 9th-grade students at Morris Area Secondary School (Minnesota) in the scientific method for well-crafted STEAM Expo research projects (2024, 2025)
- Wrote and delivered an adapted 1-hour solar system review for middle school students with disabilities at Premier Charter School (Missouri) to supplement their astronomy unit (2024)

Panelist for Awards and Fellowships Student Panel 2026

Office of Fellowships, University of Arizona

- Invited to provide advice on preparing fellowship applications, particularly the ARCS Scholar Award.

Professional Development Chair 2024 – 2026

Program to Advance Women Scientists, University of Arizona

- Organizes professional development events by organizing guest speakers (i.e., career pathways, public speaking, networking, and wellness) to improve access to academic/industry/government career pathways
- Supports the broader goals of PAWS to empower underrepresented populations in the sciences

Data Science Ambassador for the College of Science

Data Science Institute, University of Arizona

2025 – 2026

- Offers data science related consulting, training, and referrals to resources on-campus and online to support College of Science students, researchers, and educators
- Developed and taught a beginner-level hands-on Python workshop on data wrangling (pandas) and visualization (seaborn) using Google Colab

Faculty-Appointed Graduate Council Member

2023 – 2026

Treasurer (de-facto)

2024 – 2026

Project Mentor Co-Organizer

2024 – 2026

Department of Chemistry & Biochemistry, University of Arizona

- Co-designed a multi-phase graduate student peer-mentorship program, Project Mentor, launched Spring 2025
- Individually administered council funds, ensuring expenditures align with council goals; conducted purchases/reimbursements; safeguarded funds
- Organized 3 town halls, 22 coffee hours, 3 orientation panels, 1 recruitment breakfast, 1 recruitment poster, 3 professional development class sessions, 3 surveys, & 3 annual reports to foster graduate student success
- Strategized and implemented solutions to program concerns (e.g., time to graduation, recruitment, retention)

Peer Mentor

2024 – 2026

Project Mentor, Department of Chemistry & Biochemistry, University of Arizona

- Mentored one cohort of 1st year graduate students and two cohorts of 2nd year graduate students in meeting the requirements for Ph.D. candidacy
- Mentored Farinaz Soleyman, selected as the 2026 Carl S. Marvel Memorial Scholar
- Mentored Evelyn Cheng, selected for the 2026 Todd W. Rockway Early Career Award
- Mentored Zachary Bachler, a finalist for the 2026 David F. O'Brien Graduate Fellowship in Chemistry
- Delivered oral exam guidance and practice exams for Evelyn Cheng, Raushan Anjum, Anikur Rahman, Sivaraj Velusamy, Zachary Bachler, Sydney Cordova, Anna Makara, Yasaman Fazeli, and Mariposa Cox.

Astronomy Graduate School Admissions Mock Interviewer (2 students)

2025 – 2026

Científico Latino

Graduate Student Orientation Volunteer

2023 – 2026

Department of Chemistry & Biochemistry, University of Arizona

Graduate School Application Mentor

2025

Científico Latino

- Supported Lizette Rodriguez ('26, University of Texas San Antonio) in her graduate school applications. Lizette matriculates to University of Massachusetts Lowell (Physics) in Fall 2026.

Selected Participant

2025

Congressional Visit Days, American Astronomical Society, Washington, DC

Chambliss Award Judge

2025

American Astronomical Society Meeting #246

12th Grade Resume Workshop Mentor

2025

The Opportunity Network, New York

Chemistry & Biochemistry Undergraduate Poster Fair Judge

2024

Department of Chemistry & Biochemistry, University of Arizona

Travel Grant Reviewer (15 proposals)	2024 – 2025
Graduate & Professional Student Council, University of Arizona	
STEM Ambassador and Kitt Peak 12m Telescope Observatory Tour Leader	2024
Summer Space Camp by SolarSpace Company; University of Arizona; Khalifa University	
Citywide Star Party at Haverford College Host	2020
West Philly STEM-A-THON Volunteer	2018
Science in Clark Park Volunteer	2018
Philadelphia Science Festival, Philadelphia, PA; Saturdays of Service Club, Bryn Mawr College	
Co-President	2020 – 2021
Social Media & Graphic Design Chair	2019 – 2021
Volunteer	2018 – 2021
Public Observing Outreach Group, Haverford College	
Led public observing events (science demonstrations, talks/Q&A sessions, observing programs and tours, crafts) at the observatory for the community as well as private events for classrooms, religious groups, and scout troops; increased attendance from about 40 to over 200, promoting curiosity and astronomy education	

JOURNAL REVIEWER

Astronomy & Astrophysics

PROFESSIONAL MEMBERSHIPS

American Physical Society	2020 – Present
American Astronomical Society	2021 – Present
Laboratory Astrophysics Division of the American Astronomical Society	2023 – Present
American Chemical Society	2023 – Present
Program to Advance Women Scientists , University of Arizona	2023 – Present
Association for Women in Science	2025 – Present

CERTIFICATIONS & TRAINING

Certificate of Attendance , Global Astrochemistry Lecture Series Summer Semester	2026
AAS/APS/AAPT Teaching Strategies that Work: Faculty Teaching Institute Workshop	2026
for Astronomy & Physics Educators, American Astronomical Society Meeting #247	
Peer Review Workshop Completion Certificate , American Astronomical Society Meeting #246	2025
MENTOR Institute Training (Level 2) Certificate , University of Arizona	2025
MENTOR Institute Training (Level 1) Certificate , University of Arizona	2024

SELECT MEDIA & PRESS RELEASES

1. Guest on “The Liuniverse” podcast, hosted by Dr. Charles Liu (forthcoming)	2026
2. Chambliss Award Honorable Mention feature by University of Arizona Astronomy Dept. (link)	2026
3. Guest on “Starts with a Bang!” podcast episode 124, hosted by Dr. Ethan Siegel (link)	
4. College of Science-wide Service Award announcement (link)	2025
5. Chemistry Department Teaching Assistant spotlight article (link)	2024
6. University of Arizona News feature on phosphorus at the edge of the Milky Way (link)	2023
7. <i>The Sky Is Not the Limit</i> illustrated feature on ISMS presentation (link)	2022
8. Speaker in AAS #238 Press Conference: “Molecules in Strange Places” (link)	2021